

Mapping Loyalties: Visualizing Eurovision Voting Patterns Across Time

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Abstract — We present an interactive visualization of Eurovision Song Contest voting patterns across seven decades (1957–2025). The tool combines four linked views: an adjacency matrix, a geographic choropleth map, a force-directed network, and a pair timeline. Together they help answer three questions: which country pairs show steady mutual support, which relationships changed over time, and where professional juries and the public vote differently. All views share the same state and update together when the user changes any control (year range, vote type, region filter, edge threshold). Votes are scored using a rank-decay formula that gives much more weight to the top award than to lower-ranked votes. The map uses two color channels at once: the color family shows which region a country belongs to, and the brightness shows how many votes it receives on average. The timeline can show either a single country pair or a comparison between two regions.

1 Introduction

The Eurovision Song Contest is one of the largest annual live TV events in the world. Every participating country submits a song and then votes for the others, creating a rich network of voting relationships that shifts from year to year. Since 2016, each country casts two separate sets of votes: one from a professional jury and one from the general public.

We built an interactive tool to help users explore these voting patterns across seven decades (1957–2025). It focuses on three questions:

1. Which country pairs show **steady mutual support** over many years?
2. Which voting **relationships changed** the most?
3. Where do **jury and public votes disagree** most strongly?

The tool uses four linked views: an adjacency matrix, a geographic map, a force-directed network, and a pair timeline [6]. The rest of this report is structured as follows. Section 2 covers background concepts and the

dataset. Section 3 describes the original project plan. Section 4 explains how the tool was built. Section 5 shows results. Section 6 evaluates the tool. Section 7 concludes.

2 Background

2.1 Graph Terminology

A *directed graph* $G = (V, E)$ has a set of nodes V and a set of directed edges E . In our case, nodes are countries ($|V| \approx 52$) and each edge $\langle u \rightarrow v \rangle$ means country u gave votes to country v in a given year. The graph is *weighted* (edges carry point values), *dynamic* (the countries and votes change each year), and *multivariate* (from 2016 onward, each edge has a jury score, a public score, and their sum).

An *adjacency matrix* is a square table where cell (i, j) shows the weight of the edge from country i to country j . It works well for dense data because there is no edge overlap; it also makes it easy to compare both directions of a relationship ($A \rightarrow B$ vs. $B \rightarrow A$) side by side [5].

2.2 Visualization Framework

We follow Munzner’s What–Why–How framework [6] to describe the design. The **What** is the data: a directed, weighted, dynamic, and multivariate network. The **Why** is what the user wants to do: find stable alliances, track change over time, and spot jury/public disagreements. The **How** is how the data is shown: four linked views, a shared state, and a controls strip.

2.3 Dataset

The data comes from the EurovisionAPI dataset [4], covering the Senior Eurovision Song Contest from 1956 to 2025. We skip 1956 (no scores recorded) and 2020 (cancelled). The “Rest of the World” group voter (WLD, added in 2023) is left out everywhere. From 2016 onward, jury votes and public votes are stored separately.

3 Project Plan

Our project looks at how countries in the Eurovision Song Contest vote for each other over time. We want to help users see which countries often support each other, which voting relationships changed across the years, and where jury votes and public votes do not match. Since the data covers many countries and many years, a single view is not enough. So our plan is to combine multiple linked views that show the same data from different angles.

3.1 Data Abstraction (What)

Following Munzner’s framework [6], our dataset can be described as a dynamic, directed and weighted network. The nodes are countries that took part in Eurovision. The edges are votes from one country to another in a specific year.

Each country has attributes. The country name and geographic region (such as Nordic or Balkans) are *categorical*. The number of participations and years hosted are *quantitative*. Each vote record has a year (ordered quantitative, 1957 to 2025), a round (categorical: final or semifinal), a vote type (categorical: jury, public, or total), and a points value (quantitative, 0 to 12). From 2016 onward, jury and public scores are stored separately, giving two parallel quantitative channels per directed edge.

We also compute a few derived values. These include the mean normalized support per country pair, how often a country gives points to another (consistency), the difference between jury and public scores, and the overall trend in scores over time.

One known limitation of the dataset is that countries with few appearances (for example San Marino or Andorra) have very little data, which can make their scores look more extreme than they really are. The pre-1975 era also used different scoring rules, so comparisons across that boundary are less reliable.

3.2 Task Abstraction (Why)

The main questions we want to answer are the following.

1. Which country pairs show steady support over many years?

This matches Munzner’s *identify* and *browse* actions, with a focus on topology and time based patterns.

2. Which voting relationships changed the most over time?

This matches *compare* and *locate*, with the main target being trends.

3. Where do jury votes and public votes disagree?

This matches *identify outliers*, with the main target being attributes.

There is also a secondary task. Users should be able to get a quick overview of the whole network by filtering the data by year range, vote type and region, and by sorting the countries in a useful order. This follows Shneiderman’s *overview first, zoom and filter, then details on demand* mantra [7]: the matrix and map provide the overview, the controls strip drives zoom and filter, and the pair timeline delivers details on demand once a pair is selected.

3.3 Visualization Idioms (How)

We plan to use four linked views.

Adjacency matrix. This will be the main view. Rows and columns represent countries, and each cell shows how strongly one country supports another. This works well for dense network data because it avoids the heavy overlap that appears in a node link graph. It also makes it easy to compare one direction with the other, such as A to B and B to A [5].

Geographic map. The map will show where countries are located and how support is distributed across space. A selected country can also show outgoing vote flows as arcs. This helps users connect voting patterns to geography.

Force directed network. This view will show the network as nodes and links. It will be useful when the user filters the data or focuses on a smaller set of stronger connections. This view is helpful for seeing clusters or blocs.

Pair timeline. When a user selects a pair of countries, a timeline will show how that relationship changes across years. It will include jury, public and total votes where available. This is the clearest way to show change over time for a single relationship.

All views will be linked. If the user selects a country or a country pair in one view, the other views will update too. We also plan to add controls for year range, vote type, region, sorting and search.

3.4 Similar Existing Visualizations

Figure 1 shows a simple sketch of the planned interface. Eurovision World voting maps [1].

This example shows how one country voted in a given year. In terms of *what*, it uses country level voting data for a single contest. In terms of *why*, it mainly supports looking up and browsing one case at a time. In terms of *how*, it uses a map based display with vote connections. It is useful for quick exploration, but it does not support long term comparison and it does not help much with jury and public vote differences.

Academic bloc voting studies.

These studies look at Eurovision as a weighted voting network across multiple years. In terms of *what*, they work with multi year aggregate network data. In terms of *why*, they mainly support identifying blocs and summarizing overall structure. In terms of *how*, they often use static matrices, heatmaps or network summaries. They are useful for analysis, but they are fixed views and they do not let the user interact with the data directly.

3.5 Contrast with Existing Work

Our planned tool is different in three main ways.

First, it is interactive across time. Users will be able to move across years and inspect how a relationship grows, weakens or changes. This is harder to do in static academic figures and in single year web pages.

Second, it treats jury votes and public votes as separate but related parts of the data. This matters because the two groups do not always vote in the same way. Showing this difference clearly is one of the main goals of our design.

Third, our design connects several views instead of relying on only one. The matrix gives a strong overview, the map adds geographic context, the network view shows clusters and the timeline explains change across time. Together, these views support a wider set of tasks than the existing examples.

4 Methodology

4.1 Data Pipeline

The raw data comes from the EurovisionAPI dataset [4], a folder-per-year collection of JSON files covering the Senior Eurovision Song Contest from 1956 to 2025. An offline preprocessing script (Node.js, not deployed to the browser) walks every year folder, skips 1956 (no scores recorded) and 2020 (cancelled), and emits one record per voter, recipient, year, round, and vote type.

From 1975 onward the scoring table is fixed: 1, 2, 3, 4, 5, 6, 7, 8, 10, 12. Instead of dividing points by the maximum, we map each score to its rank and apply a decay:

$$\text{normalized} = 0.75^{r-1} \quad (1)$$

where r is the rank (12 points = rank 1, 10 points = rank 2, and so on). This means the top award is worth 1.0 and each lower rank is worth 75% of the one above it. A 12-point vote therefore counts much more than a 10-point vote, which better reflects how fans treat the top award. Before 1975 the scoring rules varied, so we fall back to dividing points by the highest score awarded that year. The full dataset has around 75 000 vote records.

We precompute four summary buckets: 2016–2025 for total, jury, and public votes; and 1957–2025 for total votes only. If the user picks any other year range, the browser recalculates on the fly from the raw data, which is fast enough at this scale.

Voting bloc detection. To group countries into voting blocs, we run weighted Louvain community detection [2] on the average-vote graph for 2016–2025. Each country gets a `cluster_id` which the adjacency matrix uses in its *By cluster* sort mode.

4.2 Frontend Architecture

The tool is a plain static web page: HTML, CSS, and vanilla JavaScript with no build step or server needed. D3.js v7 [3] draws all the charts; topojson-client handles the map data.

All four views read from a single shared state object (`js/state.js`). When the user changes any control (year range, vote type, region filter), the state updates and every view redraws automatically. The main data functions (`getFilteredPairs` and `getCountrySummary`) cache their results by year range, vote type, and round, so typing in the search box or clicking a cell does not trigger a full data rescan. The three research question buttons in the page header (Q1, Q2, Q3) apply curated presets to all controls at once, along with a table under the Legend section so a user can immediately see the answer to each question without adjusting sliders manually.

4.3 View Implementation

View A: Adjacency matrix. Rows are voter countries; columns are recipients. Cell color encodes mean normalized support via `d3.interpolateBlues`, scaled

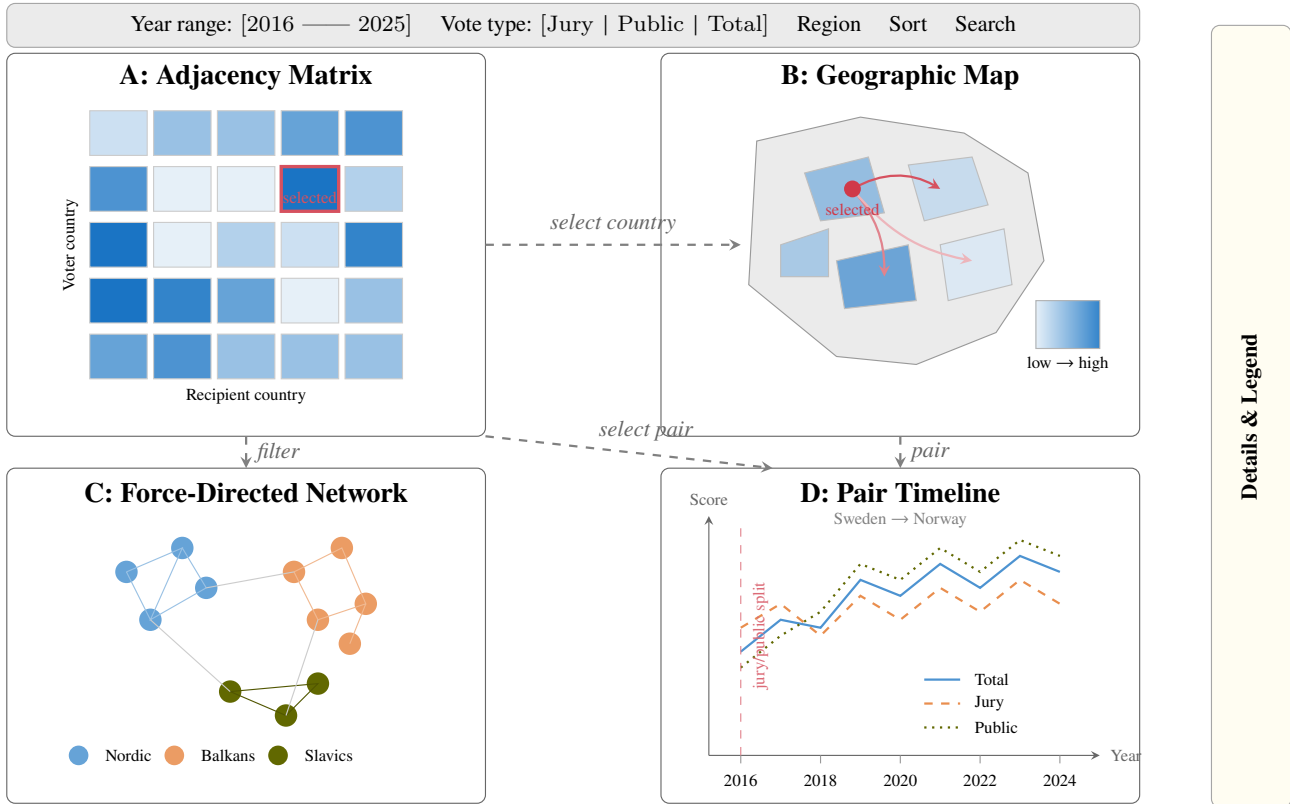


Figure 1 Design sketch of the four-view linked interface. Panel A shows the adjacency matrix heatmap. Panel B shows the geographic choropleth with arc overlays. Panel C shows the force-directed network with color-coded blocs. Panel D shows the pair timeline with jury, public and total vote lines. All views are linked through a shared state driven by the control strip at the top.

to the data maximum within the selected filters so the full color range is always used. Five sort modes are available: alphabetical, by geographic region, by average votes given, by average votes received, and by cluster. Cluster and region modes draw divider lines between groups. Clicking a cell sets `selectedPair` in shared state, which drives the pair timeline.

View B: Geographic map. The map uses a Natural Earth projection fitted to cover Europe, the Caucasus, Israel, and Morocco. Each country is colored using two channels at once. The hue channel encodes the geographic region (Nordic countries appear in blue, Balkans in warm grey, Western Europe in teal-green, and so on). The luminance channel encodes the average votes received: darker means more votes, lighter means fewer. Australia has been a participant since 2015 and is off-screen by default, but can be reached by panning. Country polygons are clipped to the screen so that overseas territories (for example French Guiana) do not create hidden clickable areas that would accidentally select the wrong country. Arc positions are calculated from capital coordinates

stored in `nodes.json`, not from geometric centroids, because countries with overseas territories have unreliable centroids. Small countries missing from the map data (Andorra, Malta, Monaco, San Marino) appear as small labeled dots using the same color scale. All drawn elements (country borders, dots, arc lines, and arc endpoint markers) shrink proportionally when zoomed in and grow when zoomed out, so they always stay the same size on screen. When a country is selected, up to five arcs going out (teal solid lines, sends votes to) and five arcs coming in (purple dashed lines, receives from) are drawn. If a country both sends to and receives from the same partner, the two arcs curve in opposite directions so neither is hidden.

View C: Force-directed network. Each country is a node, colored by region (same colors as the map) and sized by average votes received; a legend in the corner explains the sizes. Edges are drawn with arrows for all country pairs above the threshold set by the slider. The layout uses a soft pull toward approximate geographic positions, so regional groups tend to cluster together without being forced into a fixed shape. Nodes can be

dragged to a fixed position; double-clicking releases them.

View D: Pair timeline. The timeline has two modes. In single-pair mode (triggered by clicking a matrix cell), it shows three lines for the selected country pair across all years: total (solid), jury (blue dashed), and public (orange dashed). A shaded area marks the years before 2016 when only the combined total was recorded; vertical lines mark 1975 (rule change), 2016 (jury/public split), and 2020 (cancelled). In group comparison mode (triggered by picking two regions in the controls), the timeline shows the average votes from Region A to Region B and from Region B to Region A, so users can see whether the relationship is one-sided or mutual.

4.4 Design Alternatives Considered

A node-link diagram was the first candidate for the overview view. We dropped it because 52 countries and thousands of edges produce an unreadable hairball without heavy filtering. The adjacency matrix shows all pairs at once with no edge overlap and makes both directions of a relationship directly comparable. A line chart was chosen over a bar chart because it makes temporal trends and year-to-year changes easier to follow across decades.

For the network layout, we considered fixed geographic coordinates. European countries sit too close together for that to work without heavy overlap. A soft position pull keeps the rough geography visible while still spreading nodes far enough apart to read edges.

5 Results

5.1 Prototype Design and Evolution

All four views are built and working. The matrix has five sort modes, group divider lines, and sparkline tooltips. The map gained a bivariate color encoding, micro-state pin markers, a viewport clip to prevent overseas territories from interfering with clicks, and a zoom slider. The network adds soft geographic anchoring, drag-to-pin, and a size legend. The timeline gained a region comparison mode after reviewer feedback. The scoring formula was changed from a simple division to rank-decay (0.75^{r-1}) to give more weight to the top award.

5.2 Supported Tasks

All three research questions are supported by all four views.

- **Q1 (persistent support):** The adjacency matrix with cluster sort and the details panel both expose top-ally relationships over the selected year range.
- **Q2 (changing relationships):** The pair timeline shows per-year trajectories; the region-group mode extends this to bloc-level comparisons.
- **Q3 (jury/public divergence):** Toggling between *Jury* and *Televote* in the vote type control changes the matrix cell colors; the pair timeline overlays all three series simultaneously.

5.3 Illustrative Observations

Below are some patterns visible in the default view (2016–2025, finals, total vote).

Regional voting blocs (Q1). The cluster sort of the adjacency matrix reveals two visually distinct bright blocks: one covering Nordic countries (Sweden, Norway, Denmark, Finland, Iceland) and one covering Balkan countries (Serbia, Croatia, Bosnia and Herzegovina, Montenegro, Slovenia). The mean normalized score across Nordic-to-Nordic pairs is approximately 33%, more than double the global pair mean of 15%. The strongest individual connections within the Nordic group are Denmark to Sweden (66%), Norway to Sweden (62%), and Iceland to Sweden (57%). Balkan pairs show a similar internal mean of around 34%.

The single strongest sustained pair in the entire dataset, however, is Cyprus to Greece at 95% over 6 years, with Greece to Cyprus at 89% over 7 years. These two countries belong to different geographic regions in the dataset (Cyprus is classified as Levant, Greece as Balkans), so they do not appear inside the same bloc block in the cluster sort, but their individual cell is the brightest in the whole matrix. This is the clearest example of a bilateral relationship that goes far beyond regional voting.

Shifting relationships (Q2). Portugal to Ukraine rose from near-zero contact before 2016 to a 42% mean in the modern era. Switzerland to United Kingdom dropped from a 30% historic mean over 63 years to 8% today. Both shifts are visible in the pair timeline when the year range is extended to full history.

Jury and public divergence (Q3). Toggling between jury and televote in the matrix reveals pairs that

change significantly in brightness. Several cells that appear bright in one mode are near zero in the other. The pair timeline overlays both series so users can see the divergence year by year for any selected pair.

Screenshots. Figure 2 shows the full landing page with all four views visible at once. Figure 3 shows the adjacency matrix in cluster sort. Figure 4 shows the geographic map with a selected country and arc overlays. Figure 5 shows the pair timeline for an illustrative country pair. Figure 6 shows the force-directed network with the edge threshold applied. Figure 7 shows the pair timeline in region comparison mode, with the Balkans and British Isles selected.

5.4 Plans Going Forward

All four views are working. The following improvements are planned for the final version.

An animated year slider on the map would let users watch voting patterns shift over time without moving the slider by hand. The consistency and trend fields are currently only available for four precomputed year ranges; we plan to compute them for any custom range. A fullscreen button on each view panel would let users expand one view to fill the whole screen, which is useful for the adjacency matrix since it is the main analytical view and can be hard to read at 25% of the screen. The map currently shows the whole world fitted to Europe by default; we plan to clip it to Europe only and add a separate button to jump to Australia, which is the only non-European participant and does not need to be visible by default.

6 Evaluation

6.1 Informal Review

The tool was reviewed informally during development. Three problems were identified. First, the single red-to-orange choropleth made regions hard to tell apart. Second, the arc colors (originally blue and orange) were difficult to read at small sizes. Third, clicking France showed an arc starting near Spain because the arc origin used the geometric centroid of France's full polygon, which includes French Guiana and sits far south-west of Paris.

We made three changes in response. The choropleth became a bivariate encoding with hue for region and luminance for score. Arc colors became teal solid for outgoing and purple dashed for incoming. Arc origins switched to manually stored capital coordinates.

6.2 Self-Evaluation

Strengths. All three research questions can be answered with a few clicks. The Q1 shortcut button plus the mean support and min years sliders immediately surface Cyprus to Greece at 95% as the strongest consistent pair. The timeline for Switzerland to United Kingdom shows the 63-year decline to 8% in a way no static table can. Toggling jury and televote on the matrix reveals the Slovenia to Croatia gap (81%) as an instant color change, which the pair timeline then confirms year by year.

Limitations. Consistency and trend scores are zero for custom year ranges; only the four precomputed buckets have real values. Countries with few appearances (San Marino, Andorra) have sparse data, which can make their scores look more extreme than they are. The network layout is non-deterministic; regions cluster in roughly the right places but node positions shift between sessions.

7 Conclusion

We built an interactive visualization of Eurovision voting patterns covering seven decades (1957–2025). It uses four linked views: an adjacency matrix, a geographic map, a force-directed network, and a pair timeline, all connected through a shared state and a single controls strip.

All three research questions can be answered through the tool.

A few design choices turned out to be especially useful: using two color channels on the map (region and score together), giving much more weight to the top award in the scoring formula, keeping all overlaid elements the same screen size at any zoom level, and adding the region comparison mode to the timeline so users can compare whole groups rather than just single pairs.

Voting blocs detected by the Louvain algorithm [2] match geographic and cultural patterns, which confirms earlier research while also making the blocs easy to explore interactively. Next steps include an animated year slider on the map and computing trend and consistency scores for any custom year range the user picks.

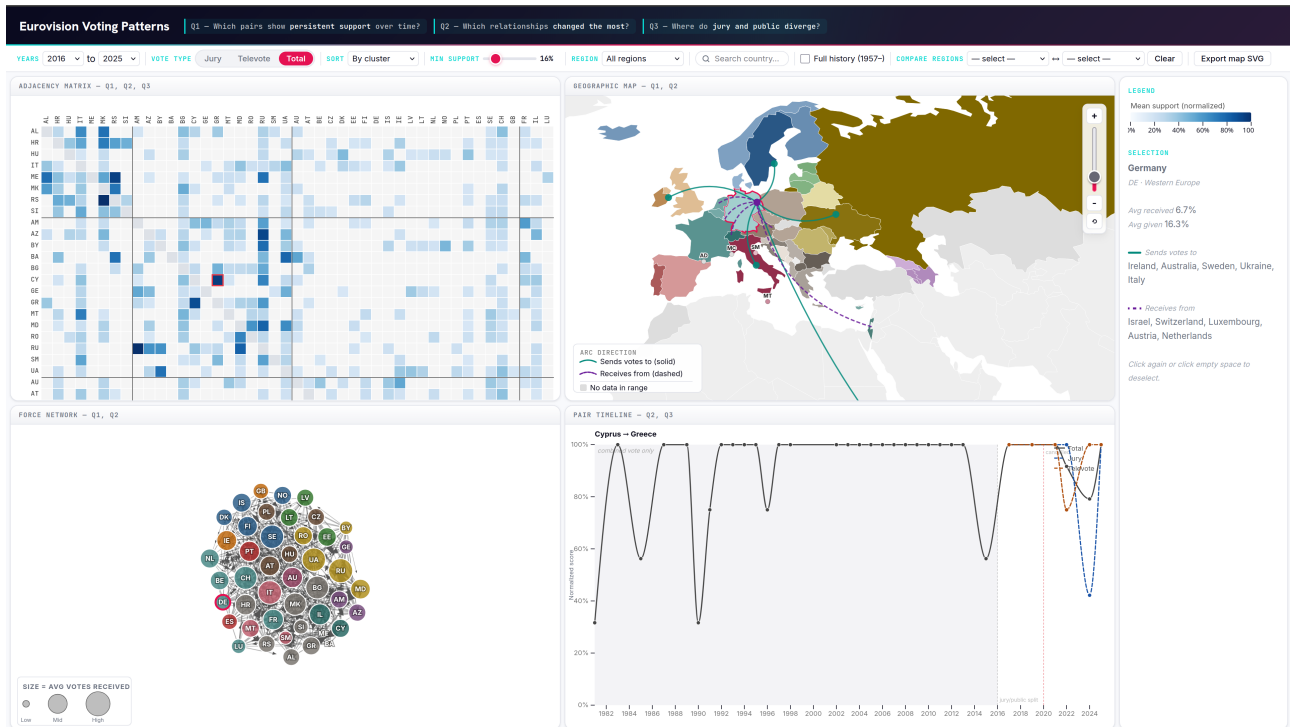


Figure 2 Full landing page of the tool showing all four views at once. The controls strip at the top (year range, vote type, sort, region, search) drives all views simultaneously. Top-left: adjacency matrix. Top-right: geographic map. Bottom-left: force-directed network. Bottom-right: pair timeline. The details panel on the right shows the selected country or pair.

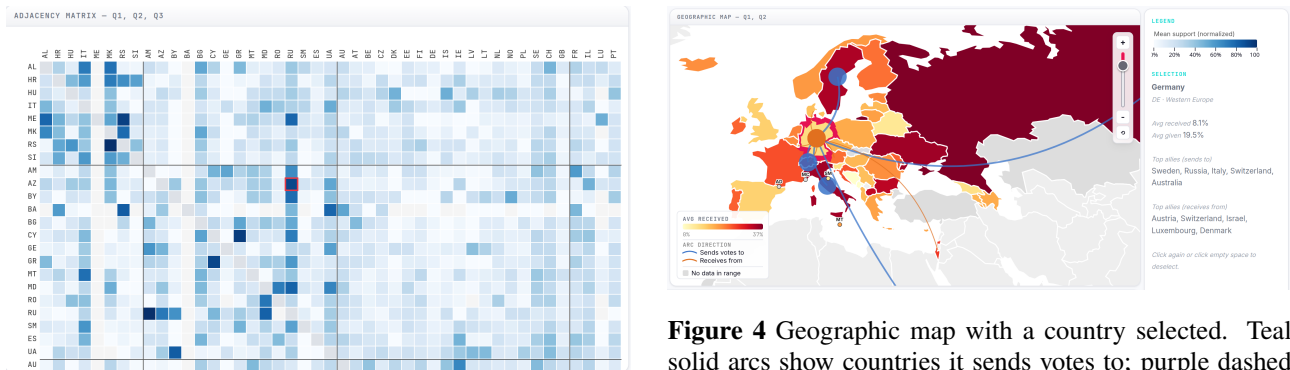


Figure 3 Adjacency matrix (cluster sort, 2016–2025, total vote). Bright diagonal blocks reveal regional voting clusters. The controls strip at the top drives all four views.

Figure 4 Geographic map with a country selected. Teal solid arcs show countries it sends votes to; purple dashed arcs show countries that send votes to it. Mutual pairs curve in opposite directions so both arcs are visible. Color encodes region (hue) and average votes received (lightness).

AI Tools Disclosure

This section is included in accordance with the course academic integrity policy.

Starting codebase. The project was not built from scratch. A working base implementation was made using several websites and tutorials examples. This included the four-view layout, the shared state and publish/subscribe architecture, the D3 rendering for all four views, the preprocessing pipeline, and the ini-

tial data files. The core architecture and design decisions were therefore inherited rather than independently conceived.

Use of AI tools in the code. AI was used to optimise and improve the existing codebase, not to create it. Specific contributions made with AI assistance include suggestions for: fixing the France arc centroid bug (overseas territories shifting the geometric centre), optimizing the matrix render to use cheap recolor and restroke paths instead of full rebuilds; Google Stitch

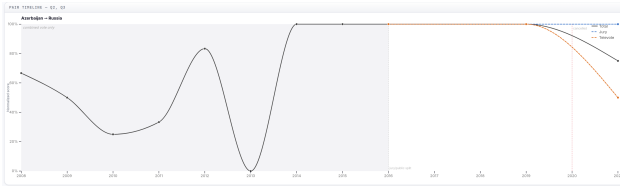


Figure 5 Pair timeline showing total (solid), jury (blue dashed), and televote (orange dashed) lines. The pre-2016 background band and the 2016 annotation line mark the introduction of independent jury and public scoring.

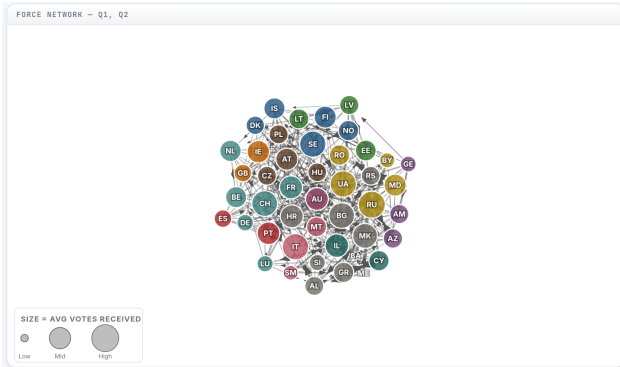


Figure 6 Force-directed network with the edge threshold applied. Node color identifies the geographic region; node size shows average votes received. Nodes can be dragged and pinned. The size legend in the corner explains the encoding.

for HTML and CSS styling to make the UI better ; and general code cleanup using Claude.

Use of AI tools in the report. AI was used to help write the report with proper grammar, structure, formatting and LaTeX syntax. The content in the report is human written. AI helped in inserting formulas, images and symbols in the right place.

Summary. AI tools were used as a coding suggestions and writing assistant on top of an existing base to help improve them. The research questions, visualization design decisions, and the overall project concept originated from the course and from the prior exercise and course work, not from AI generation.

References

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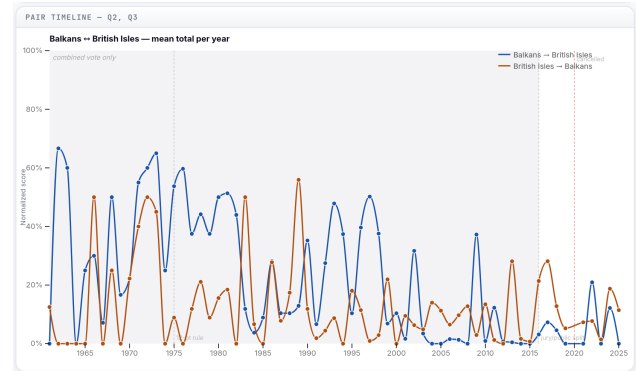


Figure 7 Pair timeline in region comparison mode: Balkans versus British Isles (2016–2025, total vote). The two lines show the average normalized votes from Balkans to British Isles and from British Isles to Balkans, making it easy to see whether the relationship is one-sided or mutual.

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